Cybersecurity Incident Report: Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.
The network traffic analysis identified an issue with DNS resolution involving UDP port 53. The captured traffic showed that DNS requests were sent using UDP to port 53, but the responses included ICMP error messages stating "udp port 53 unreachable". UDP port 53 is commonly used by the Domain Name System (DNS) to handle domain name queries and responses. DNS is responsible for translating domain names into IP addresses so that devices can locate and connect to websites. The most likely issue is that the DNS service was unavailable or not accepting requests on UDP port 53. As a result, the browser could not obtain the IP address required to access the website and establish an HTTPS connection.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.
Time of incident: 13:24:32 **Explain how the IT team became aware of the incident** Several customers reported that they were unable to access the client company's website, www.yummyrecipesforme.com. After attempting to load the page, they received a "destination port unreachable" error message. **Explain the actions taken by the IT department to investigate the incident** To investigate the issue, the analyst used the network analysis tool `tcpdump` to capture and examine packets exchanged between the computer and the DNS

server.

The analyst identified outgoing DNS requests sent using UDP to the DNS server at IP address `203.0.113.2`.

The captured traffic also showed repeated ICMP error messages stating `udp port 53 unreachable`. This indicated that the DNS requests were not being accepted on the destination port.

By examining the network traffic, the analyst was able to identify a problem affecting DNS communication and determine why the website could not be accessed.

Note key findings of the IT department's investigation (e.g. details related to the port affected, DNS server, etc.)

The investigation identified the following key findings:

- **Affected service:** Domain Name System (DNS).
- **Affected port:** UDP port 53.
- **DNS server IP address:** 203.0.113.2.
- **Protocols involved:** UDP and ICMP.
- **Error message:** "UDP port 53 unreachable".
- **Impact:** DNS requests were unsuccessful, preventing the computer from resolving the website's domain name into an IP address.

Note a likely cause of the incident

A likely cause of the incident was that the DNS service on the server was unavailable or not listening on UDP port 53. Another possible cause is that a firewall or network configuration was preventing DNS traffic from reaching the service.

This prevented successful domain name resolution, meaning the browser could not obtain the IP address needed to connect to www.yummyrecipesforme.com. Consequently, customers were unable to access the website.